3.11

ADULTS

Genital and Sexually Transmitted Infections

General Measures

1. Accurate risk assessment, education and counseling of persons at risk regarding ways to avoid STIs through changes in sexual behaviors and use of recommended prevention services.
2. All persons who receive a diagnosis of any STIs should be offered to be screened for other STIs including HIV and viral hepatitis.
3. Recommended preventive measures including HIV Pre-Exposure prophylaxis (PrEP) and vaccinations (HBV, HAV, HPV) should be discussed with the patient.
4. STI is a notifiable disease. Contact tracing needed. All partners within the past 3 to 6 months should be screened and treated epidemiologically.
5. Advise no sexual contact for 7 days after treatment is administered.

CHAPTER 3.11	GENITAL AND SEXUALLY TRANSMITTED INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Syphilis			
Early syphilis Infection of less than 2 years duration ***(Primary, secondary and***	Benzathine Penicillin 2.4 MU IM single dose	**Penicillin Allergy:** Doxycycline 100 mg PO q12H for 14 days	• Penicillin based regimens are preferred treatment for syphilis. • If allergic to penicillin, please refer to ID physician.

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early latent syphilis) *Treponema Pallidum*			
Late latent syphilis Infection of more than 2 years duration or unknown	Benzathine Penicillin 2.4 MU IM weekly for 3 weeks	**Penicillin Allergy:** Doxycycline 100 mg PO q12H for 28 days	• If patient defaults Benzathine Penicillin by more than 2 weeks in between the weekly doses, the whole regimen needs to be restarted. • If allergic to penicillin, please refer to ID physician.
Neurosyphilis	Benzylpenicillin 4 MU IV q4H for 14 days	**Penicillin Allergy:** *Limited data on the effectiveness of non-penicillin regimen* *Consider penicillin desensitisation and referral to ID physician* Ceftriaxone 2 g IV/IM q24H for 14 days (If no anaphylaxis to penicillin)	• **Indications for lumbar puncture:** • Patients with neurological and / or ocular symptoms or signs • Non-treponemal test titres do not decrease by fourfold within 12 months of therapy • CSF VDRL - 50% sensitivity but high specificity. • A negative CSF VDRL does not exclude neurosyphilis. • Limited data on the effectiveness of extended course of high dose oral doxycycline for neurosyphilis. Consider ID referral if needed.
Syphilis in pregnancy Primary, secondary, and	**1st and 2nd Trimester:** Benzathine Penicillin 2.4	**Penicillin Allergy:** No proven alternative	• Doxycycline and tetracycline are contraindicated in

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early latent	MU IM single dose **3rd Trimester:** Benzathine Penicillin 2.4 MU IM weekly for 2 weeks (Day 1 & 8)	therapy, consider penicillin desensitization. **If unable to do so, alternative:** Ceftriaxone 500 mg IM q24H for 10 days	pregnancy. • Avoid use of macrolides in early pregnancy as it is associated with increased risk of treatment failure. • When syphilis is diagnosed during the second half of pregnancy, management should include a sonographic foetal evaluation for congenital syphilis. • Pregnant women who have delays in any therapy dose >9 days between doses should repeat the full course of therapy.
Late latent syphilis and neurosyphilis	Treat as non-pregnant patients		
Gonorrhoea *Neisseria gonorrhoea*			
Uncomplicated genital, ano-rectal and pharyngeal infection	Ceftriaxone 500 mg IM single dose (if weight <150 kg) **OR** 1 g IM single dose (if weight ≥150 kg) ± Doxycycline 100 mg PO q12H for 7 days (if Chlamydia not excluded)	**Penicillin Allergy:** Gentamicin 240 mg IM single dose + Azithromycin 2 g PO single dose **Pregnancy and breastfeeding:** Treat as non-pregnant	• Please send gonorrhoea culture and antimicrobial susceptibility testing prior to antibiotic treatment of all patients with nucleic acid amplification testing (NAAT) positive gonorrhoea. • Test of cure is not recommended except for pharyngeal NG infection. • NAAT test should not be repeated before 4 weeks post treatment due to increased likelihood of false positive result. • Patient with suspected treatment failures should first be re-treated routinely with the initial regimen used as reinfections

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		adults	are common. However, relevant clinical specimen should be obtained for culture and sensitivity testing (preferably with simultaneous NAAT).
Gonococcal epididymitis / epididymo-orchitis	Ceftriaxone 500 mg IM single dose (if weight <150 kg) **OR** 1 g IM single dose (if weight ≥150 kg) ± Doxycycline 100 mg PO q12H for 10-14 days		
Disseminated gonococcal infection	Ceftriaxone 1 g IV/IM q24H	Cefotaxime 1 g IV q8H	• For hospital admission. • May be switched to Ciprofloxacin 500 mg PO q12H 24-48hrs after symptoms improve and if susceptible. • Duration of treatment: 7 days.
Gonococcal conjunctivitis	Ceftriaxone 1 g IM single dose	**Penicillin Allergy:** Azithromycin 2 g PO single dose + Doxycycline 100 mg PO	• A 3-day regimen is recommended as cornea may be involved and is relatively avascular. • The eye should be irrigated with saline and water.

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		q12H for 1 week + Ciprofloxacin 250 mg PO q24H for 3 days	
Chlamydia *Chlamydia trachomatis serovars D to K*			
Uncomplicated (urogenital, pharyngeal, and rectal infection)	Doxycycline 100 mg PO q12H for 7 days	Azithromycin 1 g PO single dose	• Test of cure is not routinely recommended.
Chlamydia in pregnancy	Azithromycin 1 g PO single dose	Amoxicillin 500 mg PO q8H for 7 days	• Doxycycline is contraindicated in pregnancy. • Test of cure (TOC) is recommended in pregnancy, poor compliance, and persistent symptoms. Test of cure should be performed 4-6 weeks after treatment with NAAT. Retesting after 3 months of treatment is also recommended.
Mycoplasma genitalium			
Uncomplicated urogenital infection (urethritis cervicitis)	Doxycycline 100 mg PO q12H for 7 days **Followed by,** Moxifloxacin 400 mg PO		• Please refer to ID if symptoms persist despite compliance to treatment.

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	q24H		
Non-specific urethritis (NSU) / Non-gonococcal urethritis (NGU)			
First episode of non-gonococcal urethritis (NGU)	Doxycycline 100 mg PO q12H for 7 days	Azithromycin 1 g PO single dose	• Most common cause of recurrent or persistent NGU is *Mycoplasma genitalium*. • Follow-up is recommended after 2-3 weeks. • Those with persistent or recurrent NGU after treatment for *M. genitalium* or *T. vaginalis* should be referred to an infectious disease or urology specialist.
Recurrent and persistent non-gonococcal urethritis (NGU)	To assess compliance and potential re-exposure. To extend diagnostic testing to include *M. genitalium* and *T. Vaginalis* in addition to *Chlamydia trachomatis* and *Neisseria gonorrhoeae* (CT/NG).		
	If *T. Vaginalis* detected: Metronidazole 2 g PO single dose Otherwise consider treating *M. genitalium* as above.		
Chancroid			
Haemophilus ducreyi	Azithromycin 1 g PO single dose **OR** Ceftriaxone 250 mg IM		• Needles aspirate fluctuant buboes from adjacent healthy skin – less commonly seen nowadays.

Infections and Likely Organisms	Suggested Treatment		Notes
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	single dose **OR** Ciprofloxacin 500 mg PO q12H for 3 days **OR** Erythromycin Ethyl Succinate 800 mg PO q6H for 7 days		
Lymphogranuloma venereum (LGV)			
Chlamydia trachomatis serovars L1, L2, L3	Doxycycline 100 mg PO q12H for 21 days	Azithromycin 1 g PO weekly for 3 weeks **OR** Erythromycin Ethyl Succinate 800 mg PO q6H for 21 days	• Drainage of infected buboes may be required.
Granuloma inguinale (Donovanosis)			
Klebsiella granulomatis	Azithromycin 1 g PO weekly for 3 weeks **OR** Azithromycin 500 mg PO q24H for 7 days	Co-trimoxazole (480 mg) 2 tablets PO q12H for 3 weeks **OR** Ceftriaxone 1 g IM / IV	• Duration: minimum of 3 weeks and should be continued until the lesions have healed. • Add Gentamicin 1 mg/kg IM / IV q8H in patients whose lesions do not respond in the first few days to other agents.

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		q24H for 3 weeks **OR** Doxycycline 100 mg PO q12H for 3 weeks	
Trichomoniasis			
Trichomonas vaginalis	**Women:** Metronidazole 400 mg PO q8H for 5-7 days **Men:** Metronidazole 2 g PO single dose	**Women and men (if unable to tolerate Metronidazole):** Tinidazole 2 g PO single dose*	• High dose Metronidazole or Tinidazole is not recommended in pregnancy or breastfeeding. • Avoid alcohol (disulfiram effect) with Metronidazole. • In Metronidazole-resistant trichomoniasis, use higher doses of Metronidazole e.g., 2 g PO daily for 3 - 5 days or Clotrimazole intra-vaginal cream. • *Not available in UMMC.
Genital *herpes virus* Type-1 and 2 (HSV-1 and 2)			
First clinical episode	Acyclovir 400 mg PO q8H for 7-10 days	Valaciclovir 1 g PO q12H for 7 days*	• Treatment can be extended if healing is incomplete after 10 days of therapy
Recurrent HSV	Acyclovir 800 mg PO q12H for 5 days	Valaciclovir 500 mg PO q12H for 3 days*	• *Valaciclovir is not available in UMMC.

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	OR Acyclovir 800 mg PO q8H for 2 days **OR** Acyclovir 400 mg PO q8H for 5 days	**OR** Valaciclovir 1 g PO q24H for 5 days*	
Suppressive therapy (Indicated if ≥ 6 recurrences per year, severe, prolonged or with psychosocial problems)	Acyclovir 400 mg PO q12H	Valaciclovir 500 mg PO q24H*	• Duration: up to 1 year, then reassess. • If breakthrough recurrences occur, dosage should be increased (refer: Recurrent HSV). • *Valaciclovir is not available in UMMC.
Severe disease (complications that necessitate hospitalisation e.g., disseminated infection, pneumonitis, hepatitis or CNS complications)	Acyclovir 5-10 mg/kg IV q8H 10-14 days	Foscarnet 40-80 mg/kg IV q8H 10-14 days*	• Intravenous therapy should be considered until clinical improvement followed by oral antiviral therapy to complete > 10 days of total therapy. • *Foscarnet is not available in UMMC.
Genital herpes in	**First or second**	Valaciclovir 500 mg PO	• Do not delay treatment whilst awaiting results (HSV PCR

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pregnancy	**trimester acquisition (until 27th weeks):** Acyclovir 400 mg PO q8H for 5 days and start suppressive therapy at 36 weeks until delivery	q12H for 5 days*	recommended) • For all women who present with a genital HSV lesion anytime during pregnancy, suppressive therapy initiation at 36 weeks of gestation is recommended.
	Third trimester acquisition (from 28th weeks): Acyclovir 400 mg PO q8H for 5 days followed with suppressive therapy until delivery	Valaciclovir 500 mg PO q12H for 5 days*	
Suppressive therapy for recurrent genital herpes in pregnancy	Acyclovir 400 mg PO q8H (Treatment recommended starting at 36 weeks of gestation until delivery)		